GATE PYQs

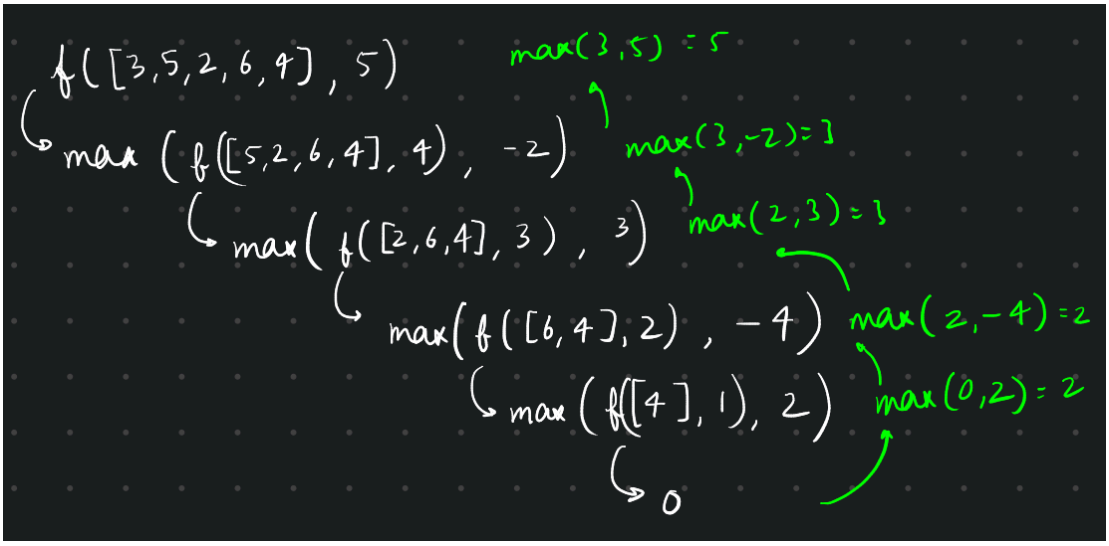
GATE 2016

```python
def f(p: list, n: int) -> int:
    if n <= 1:
        return 0
    else:
        return max(f(p[1:], n-1), p[0]-p[1])

a = [3, 5, 2, 6, 4]
print(f(a, 5))
```

What is the output of the above code snippet?

ans = 3



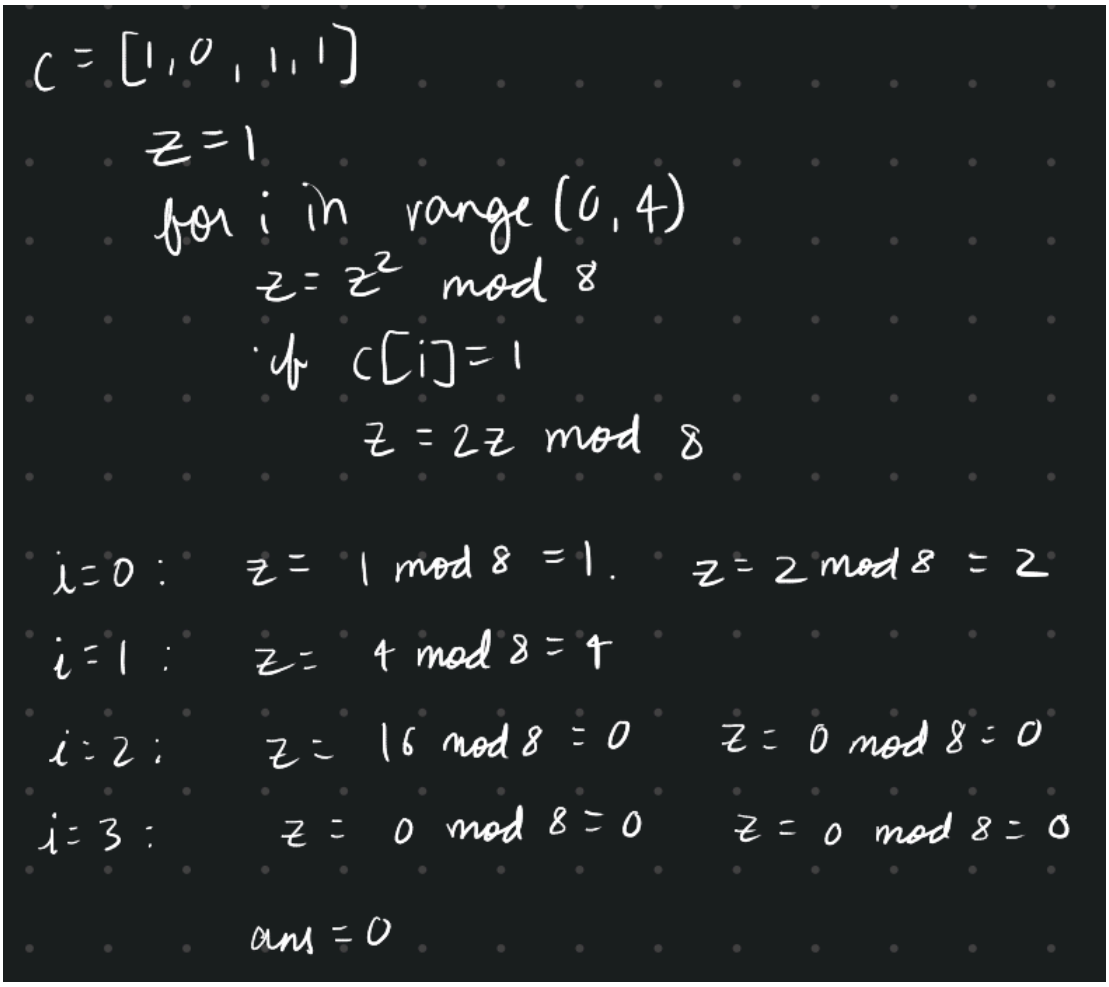
GATE 2015

Suppose c = (c[0], ..., c[k-1]) is a list of length k, where all the entries are from the set {0, 1}. For any +ve integers a and n, consider the following pseudocode.

```
DOSOMETHING(c, a, n):
    z = 1
    for i in range(0, k):
        do z = z^2 mod n
        if c[i] = 1:
            do z = z*a mod n
    return z
```

If k=4, c = (1, 0, 1, 1), a=2 and n=8, the output of DOSOMETHING(c, a, n) is _____.

ans = 0



GATE 2015

```
x = 10
def f1():
    x = 25
    x += 1
    return x

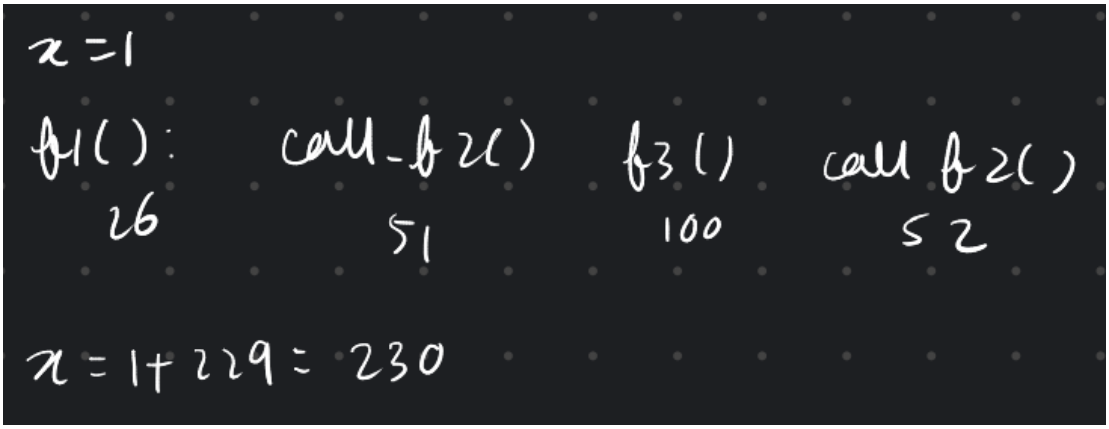
def f2():
    x = 50
    def inner_f2():
        nonlocal x
        x += 1
        return x
    return inner_f2()

def f3(y=x):
    z = y*10
    return z

x = 1
call_f2 = f2()
x += f1() + call_f2() + f3() + call_f2()
print(x)
```

What is the output of the above code snippet?

ans = 230

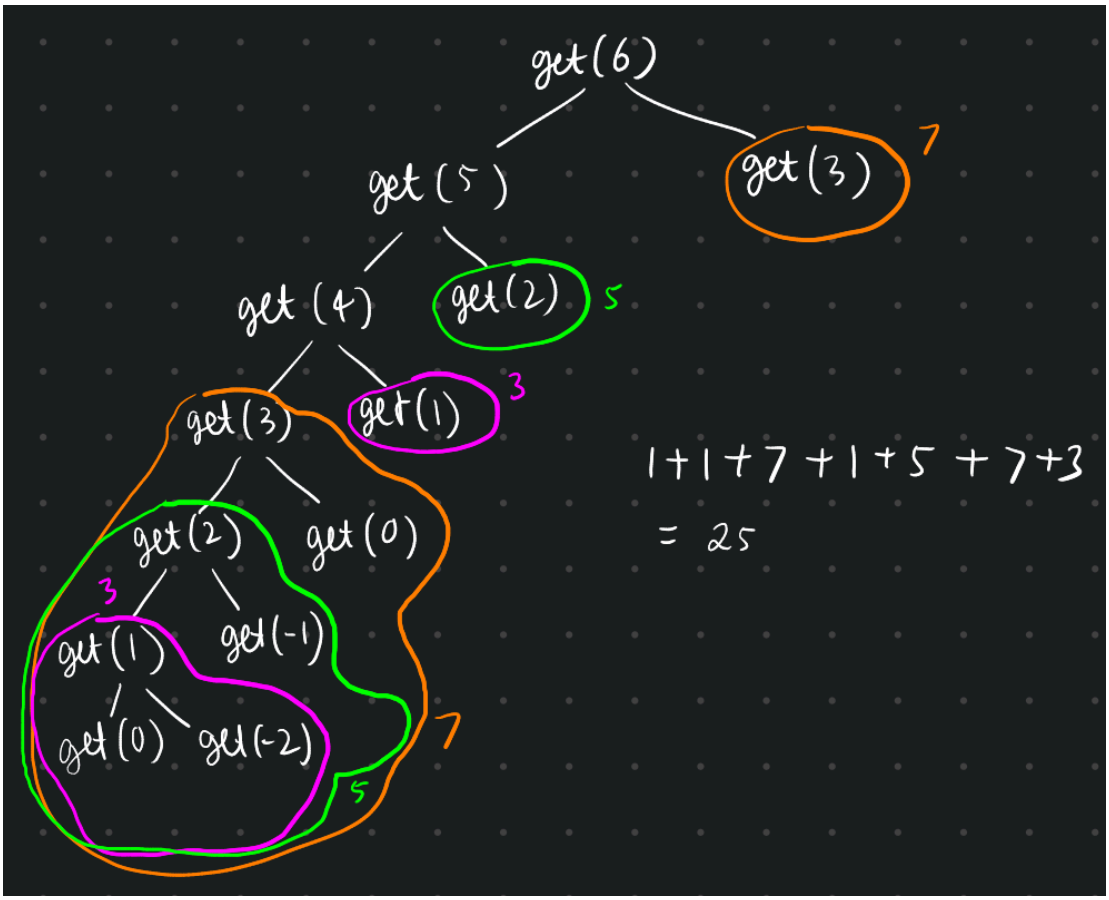


GATE 2015

```
def get(n):
    if n < 1:
        return
    else:
        get(n-1)
        get(n-3)
```

If `get(6)` is called, how many times will `get()` be invoked?

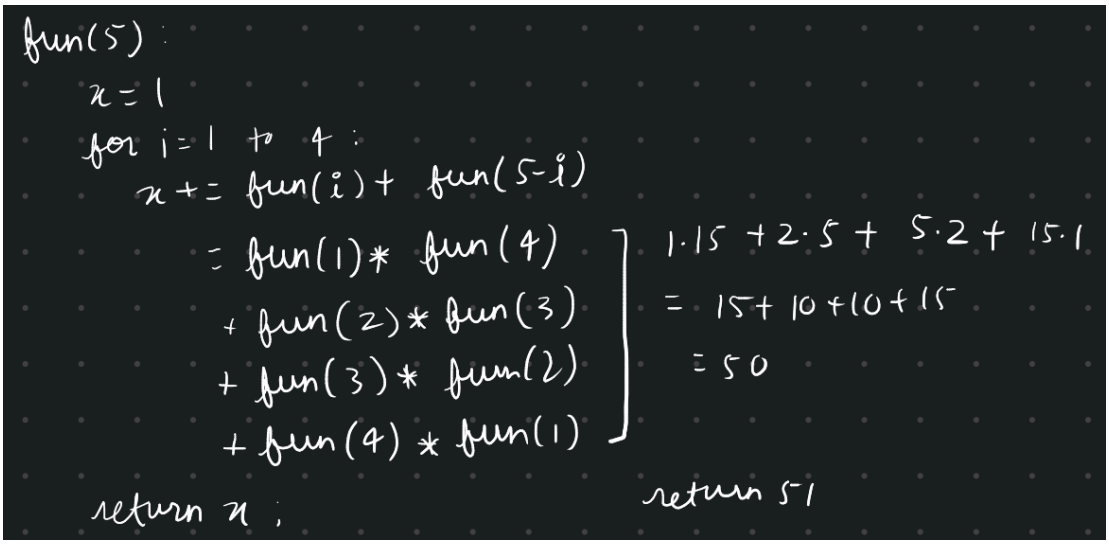
ans = 25

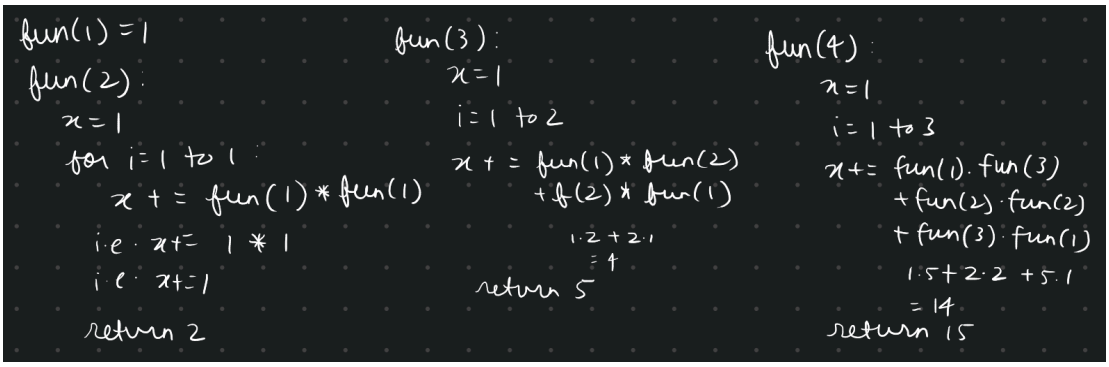


GATE 2015

```
def fun(n):
    x = 1
    if n == 1:
        return x
    else:
        for k in range(1, n):
            x += fun(k)*fun(n-k)
        return x
```

What is the value of `fun(4)` ?





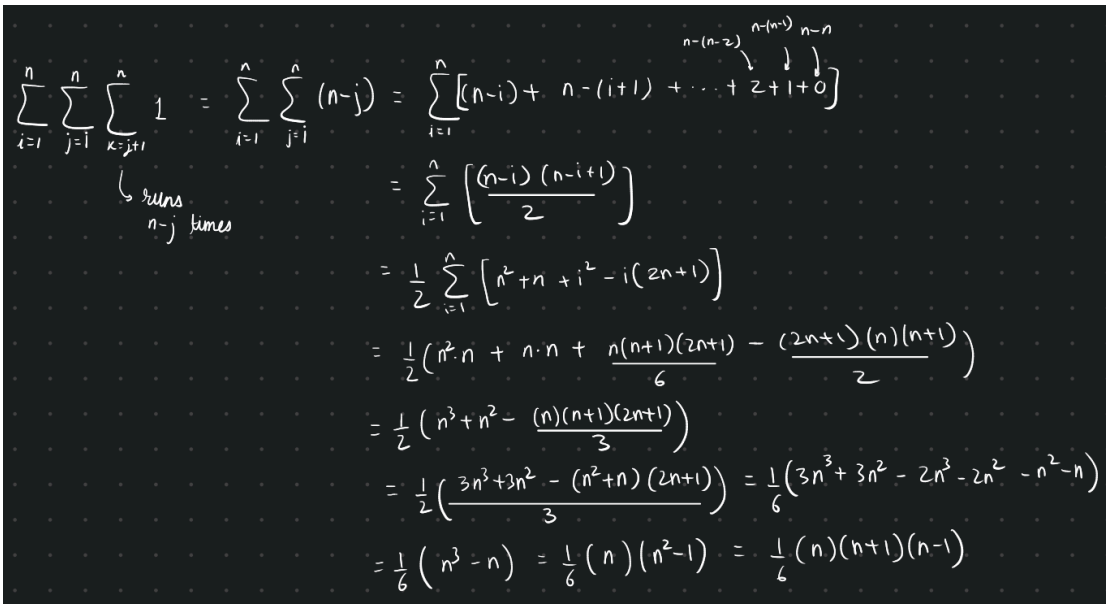
GATE 2014

Consider the following pseudocode. What is the total number of multiplications to be performed?

```
D = 2
for i = 1 to n do
    for j = i to n do
        for k = j+1 to n do
            D = D * 3
```

a. half of the product of the 3 consecutive integers
b. one-third of the product of the 3 consecutive integers
c. one-sixth of the product of the 3 consecutive integers
d. none of the above

ans = option c



GATE 2022

Unicode representation:

A B ... Z

65 66 ... 90

a b ... z

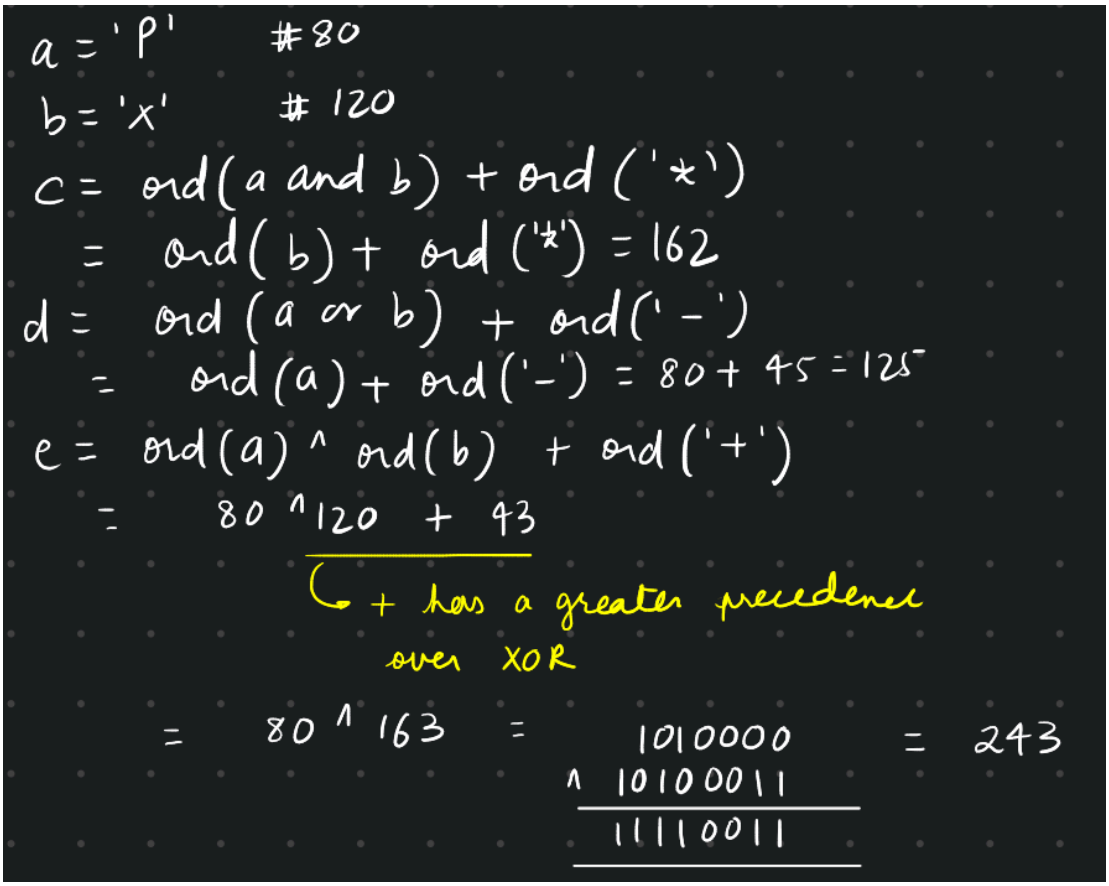
97 98 ... 122

* + -

42 43 45

```
a = 'P'
b = 'x'
c = ord(a and b) + ord('*')
d = ord(a or b) + ord('-')
e = ord(a)^ord(b) + ord('+')
print(c+d+e)
```

ans = 243


a = 'P' #80
b = 'x' #120
c = ord(a and b) + ord('*')
= ord(b) + ord('*') = 162
d = ord(a or b) + ord('-')
= ord(a) + ord('-') = 80 + 45 = 125
e = ord(a) ^ ord(b) + ord('+')
= 80 ^ 120 + 43
+ has a greater precedence over XOR
= 80 ^ 163 = 1010000 ^ 10100011 = 11110011 = 243